DATABASE MANAGEMENT SYSTEMS (ENCS204)

ASSIGNMENT → 2

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Section :- A

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Thursday.

ASSIGNMENT → 2

[SECTION-A]

Q1) fill in the blanks:-

1) The Entity-Relationship (ER) visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

2) In the Relational Model data is organised into tables, with relationships b/w entities defined through foreign key constraint.

3) The object-oriented model treats data as a collection of objects, each with unique attributes and methods, mirroring real world entities and interactions.

4) Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

[SECTION-B]

Q2) In a healthcare system, discuss how data abstraction can simplify the interaction b/w healthcare professionals and patient records.

Ans Data abstraction in a healthcare system hides complex database details and show only necessary information to users.

→ Doctors can quickly view patient medical history, test reports, and perscriptions without dealing with database structure.

→ Nurses can update patient vitals and treatment details easily.

→ Administrative staff can mange, appointments, billing and patient records.

It improves data security, because users only see the information they are authorized to access.

Q3) Explore the potential benefits and challenges associated with implementing data abstraction in the context.

Ans. Improved security → sensitive ~~da~~ patient data is protected

→ Easy data management :- Complex database structure is hidden.

→ Better efficiency :- healthcare staff can access required data quickly.

→ Reduced errors → clean and simplified interfaces help avoid mistakes.

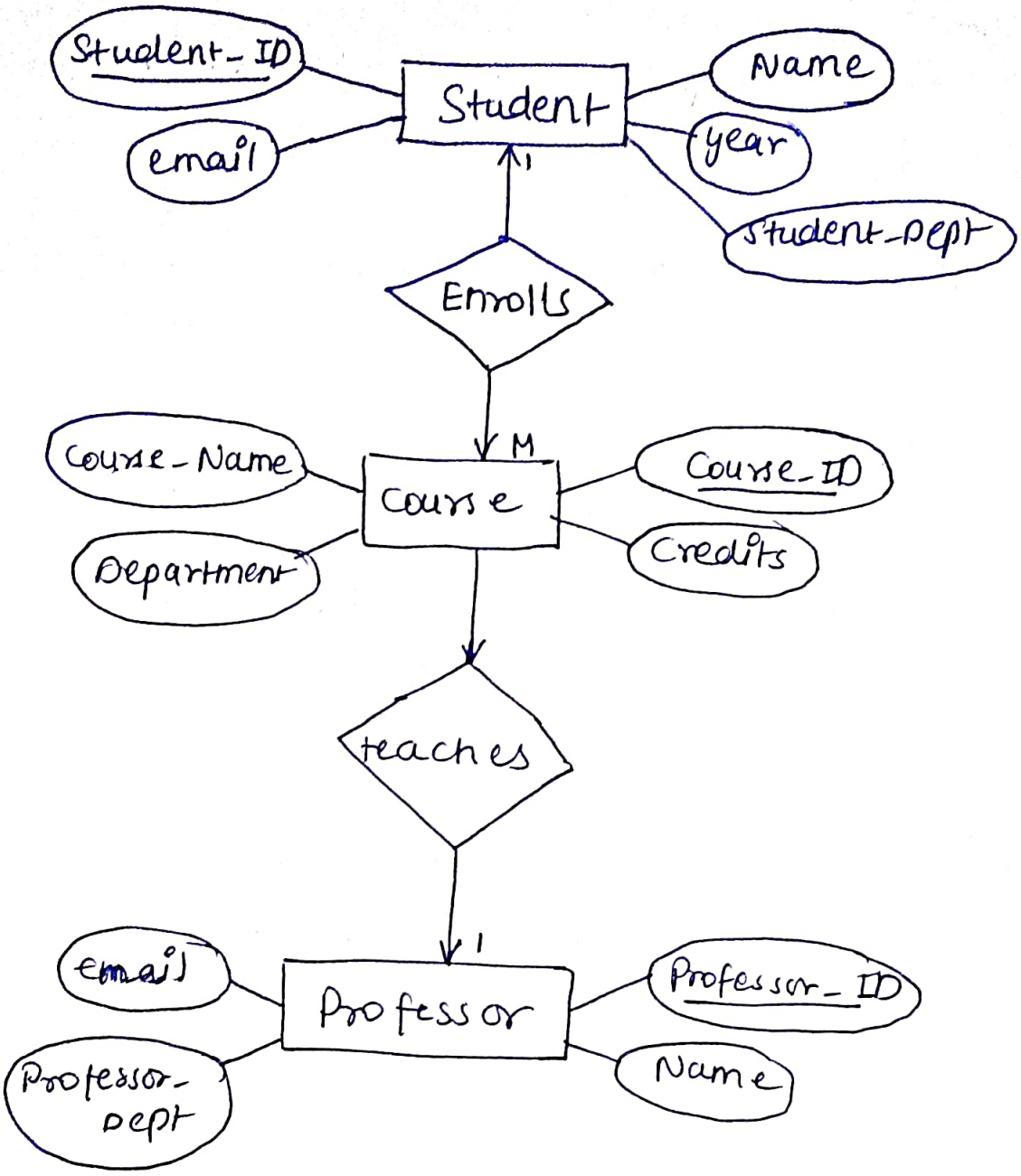
* Challenges

→ Implementation cost :- Designing abstraction layers can be expensive.

→ System complexity :- Requires careful planning and skilled developers.

→ Performance issues :- Too many abstraction layers may slow the system

→ Training requirement :- staff may need training to use the system properly.

[SECTION - C]

Q4) Design a database for a university:-